

HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE

VitalTwin

A personal digital twin for chronic-disease patients that monitors, predicts and simulates

Team: [Team name]

Members: [Member 1], [Member 2], [Member 3]

Institution: [College / Incubator name, City]

The problem: chronic disease is managed in snapshots

- **India has 100M+ people with diabetes** and ~315M with hypertension; many are undiagnosed or poorly controlled (ICMR-INDIAB, 2023).
- **Care is episodic** - a clinic visit every 3 months sees one BP reading and one HbA1c, while the disease evolves every day.
- **Wearables & CGMs produce rich data**, but nobody turns it into a personal, predictive model of the patient.
- **Interventions are trial-and-error** - doctors cannot test 'what if this patient walks 30 min and cuts salt?' before prescribing it.

Our question

Can we build a living virtual replica of each patient that stays synced with their real data, flags deterioration early, predicts long-term risk, and lets clinicians safely test interventions on the twin first?

Healthcare use case

Patient

Sees their own twin
Understands what drives risk
Plans meals & activity on the glucose twin before eating

Clinician

Remote monitoring dashboard
Prioritised critical alerts
Tests therapy plans on the twin before prescribing

Care program

Population triage by risk
Preventive outreach for pre-diabetic / hypertensive members (e.g. corporate wellness)

Example journey - Ravi, 52, pre-diabetic, hypertensive smoker

- **Twin syncs** 48 h of smartwatch, CGM and BP-cuff data and calibrates his insulin sensitivity (–29 % vs population prior).
- **Monitoring** catches a tachycardia episode, a night-time SpO₂ dip and a hypoglycaemic event as critical alerts.
- **Risk view** shows 15 % diabetes / 19 % CVD 10-yr risk; smoking and systolic BP are the top modifiable drivers.
- **What-if** a realistic plan (300 kcal deficit, 150 min/wk exercise, 2.5 g salt, quit smoking, 80 % adherence) cuts CVD risk to ~6 % in 24 weeks.

Solution: a hybrid physics + AI digital twin

Sync

Ingest multi-sensor streams
EWMA state estimation
Calibrate physiology to the patient's CGM (least squares)

Monitor

Personal baselines
IsolationForest anomalies
Clinical hard limits
De-duplicated alert feed

Predict

10-yr T2D & CVD risk
Calibrated gradient boosting
Counterfactual explanations
of modifiable drivers

Simulate

Glucose-insulin ODE (24 h)
Lifestyle & medication
projection (12-52 wks)
Scenario vs. no-change

Why hybrid? Mechanistic models give physiologically plausible, explainable simulations with little data; ML models capture population risk patterns that equations miss. Together they make a twin that can both explain and predict.

System architecture

1 · PHYSICAL PATIENT & DATA

 **Smartwatch**
HR · SpO₂ · skin temp · steps (1 min)

 **CGM sensor**
Glucose every 5 min

 **BP cuff**
Systolic / diastolic, 3×/day

 **EHR & labs**
HbA1c · lipids · history · meds

 **Patient log**
Meals · exercise · sleep

2 · INGESTION & TWIN SYNC

Stream ingestion
Resample to 5-min windows
Interpolate gaps, de-noise
pandas pipeline (twin/data.py)

State estimation
EWMA blend of EHR + sensors
Resting HR · BP · fasting glucose
(twin/sync.py)

Physiological calibration
Fit insulin sensitivity SI and
basal glucose Gb to CGM
SciPy Nelder-Mead least squares
RMSE → sensor-noise level

3 · DIGITAL TWIN CORE

Twin state store
Patient profile + calibrated parameters

Physiology engine (mechanistic)
Bergman minimal model: glucose-insulin,
meals, exercise, metformin (1-min ODE)
Lifestyle model: weight · BP · HR · LDL
· HbA1c (ADAG) week-by-week

AI / ML engine (data-driven)
Calibrated HistGradientBoosting:
10-yr T2D & CVD risk (AUC 0.87 / 0.82)
Counterfactual risk-driver explainer
IsolationForest + personal z-score
+ clinical limits → alerts

4 · APPLICATIONS (STREAMLIT)

 **Twin state**
Sync view & calibration fit

 **Live monitoring**
Vitals replay + alert feed

 **Risk drivers**
Explainable 10-yr risk

 **What-if simulator**
Compare plans over 12-52 wks

 **Glucose twin**
Plan a day: meals, walk, meds

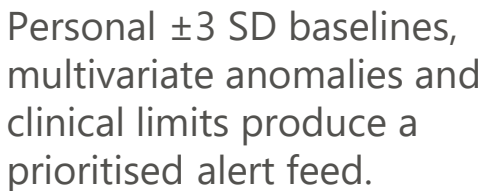
Closed loop: clinician / patient chooses an intervention on the twin → applied to the real patient → new sensor data re-syncs and re-calibrates the twin

AI / ML models & frameworks

Component	Method	Framework	Result
10-yr T2D risk	HistGradientBoosting + isotonic calibration	scikit-learn	AUC 0.868 · Brier 0.030
10-yr CVD risk	HistGradientBoosting + isotonic calibration	scikit-learn	AUC 0.818 · Brier 0.046
Risk explanation	Counterfactual: factor → healthy target	custom	Per-patient ranked drivers
Anomaly detection	IsolationForest + personal z-score + limits	scikit-learn	All injected events caught
Glucose physiology	Bergman minimal model + gut absorption	NumPy (Euler, 1 min)	Est. HbA1c within 0.3 % of labs
Twin calibration	Nelder-Mead least squares on SI, Gb	SciPy	SI recovered within ~2 %
Lifestyle projection	Effect-size model, coupled to ODE	NumPy	Weight · BP · LDL · HbA1c

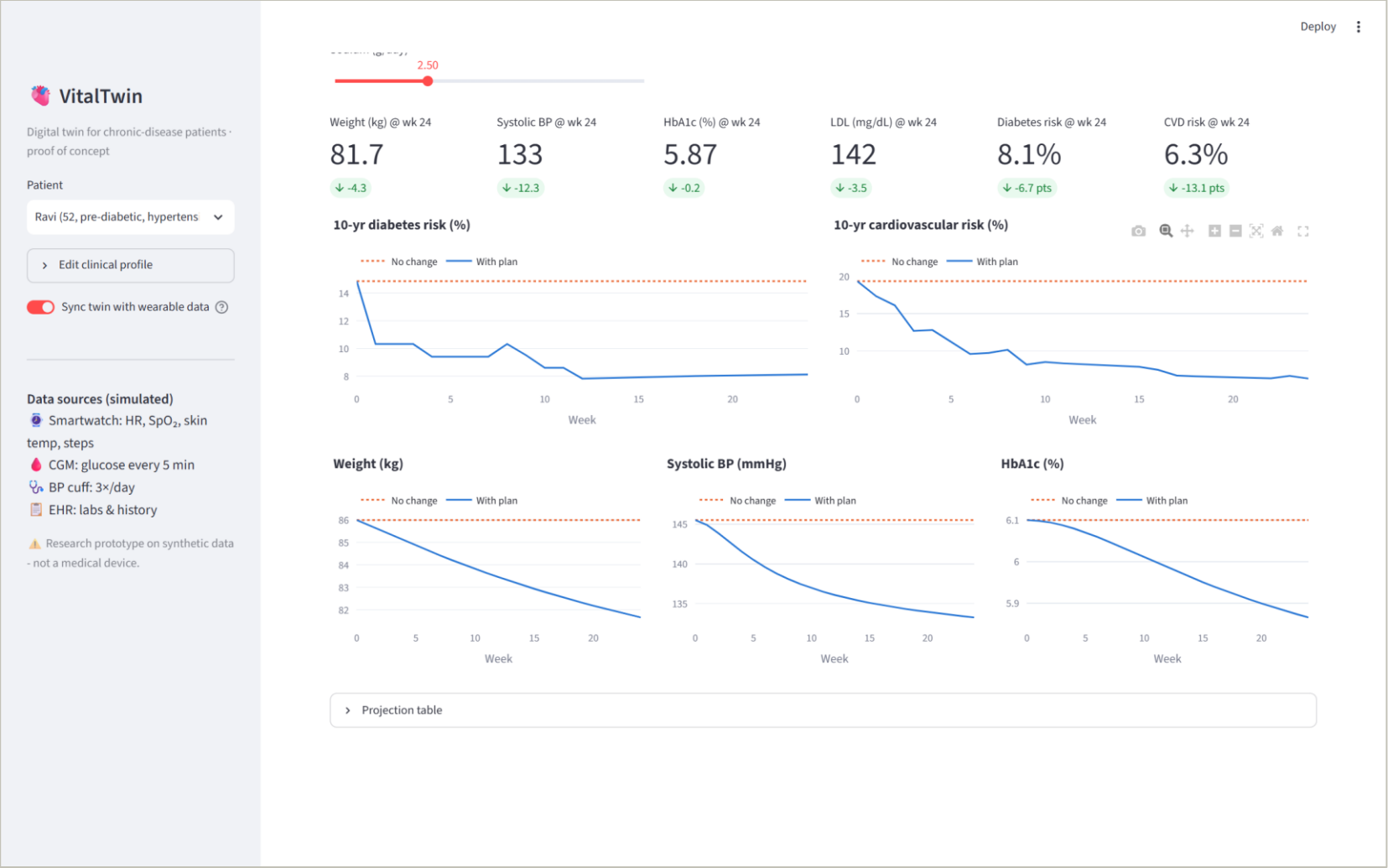
Trained on a 20,000-person synthetic cohort with FINDRISC / Framingham-inspired risk structure (80/20 stratified split). No real patient data is used; the pipeline is ready to retrain on NHANES, UK Biobank or hospital EHR data.

EHR values are blended with 48 h of sensor data; the glucose model is fitted to this patient's CGM trace.

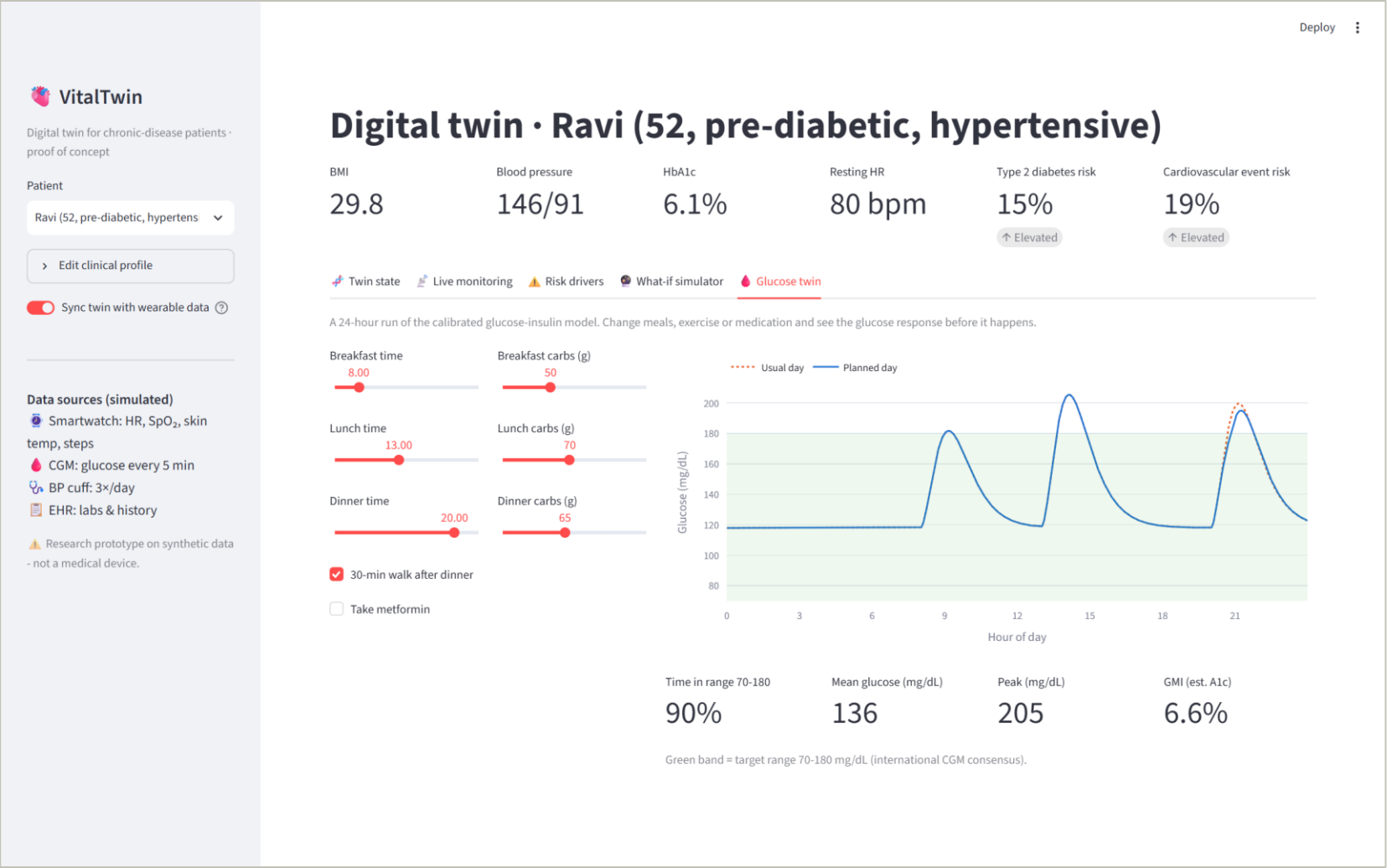


Each bar = risk reduction if that one factor reached its healthy target - actionable, not just a score.

What-if intervention simulator



Glucose twin: plan a day



Outcomes

0.87

AUC · diabetes risk

0.82

AUC · CVD risk

~2 %

error recovering hidden insulin
sensitivity

4 / 4

injected clinical events detected

- **Working end-to-end twin** - sync, monitor, predict, explain and simulate in one interactive app.
- **Physiologically grounded** - simulated HbA1c within 0.3 % of each persona's lab value before any calibration.
- **Actionable** - for the demo patient a realistic plan lowers 10-yr CVD risk from 19 % to ~6 % and diabetes risk from 15 % to ~8 % in 24 weeks.
- **Tested** - 10 automated tests cover the physiology, calibration, ML ranking and anomaly detection.

Limitations & roadmap

Current limitations

- Synthetic training cohort - risks are illustrative
- Effect sizes simplified from literature
- Simulated sensor streams, single-patient app
- Not clinically validated; not a medical device

Roadmap

- Retrain on real cohorts (NHANES, ICMR-INDIAB, EHR)
- Live device APIs: Health Connect, HealthKit, Libre
- FHIR integration & clinician multi-patient view
- Kalman / Bayesian twin updating with uncertainty bands
- Cardiac (HRV, ECG) and renal sub-models
- Prospective validation study with a partner hospital

Thank you

Team: [Team name]

Members: [Member 1], [Member 2], [Member 3]

Institution: [College / Incubator name, City]

Code: <https://github.com/AnushreeKasturi/health-digital-twin>

Demo video: [Unlisted YouTube link]

Research prototype on synthetic data. Not intended for diagnosis or treatment decisions.